

Rectifying Multimodal Variance: UAPA-HCF for Weakly Supervised Violence Detection

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Abstract

Weakly supervised multimodal violence detection aims to localize violent segments in untrimmed videos using only video-level labels. However, RGB, audio, and optical flow differ substantially in information density and noise sensitivity. Naive fusion can therefore introduce high-entropy background signals into cross-modal interactions, weakening joint representations and causing false alarms. To address this, we propose UAPA-HCF, a rectify-before-fusion framework with two components. Uncertainty-Aware Prototype Anchoring (UAPA) maintains a learnable dictionary of violence prototypes as semantic anchors and estimates snippet-level aleatoric uncertainty for auxiliary modalities. Its probabilistic gate suppresses noise-corrupted evidence while aligning reliable features with the prototype space and RGB-guided context. Holistic Consensus Fusion (HCF) then dynamically weights the corrected modalities according to their credibility, and a MIL objective converts the fused representation into segment-level violence scores. On XD-Violence, UAPA-HCF achieves state-of-the-art performance among directly comparable multimodal methods, reaching 87.65% AP under the standard protocol. These results support the effectiveness of evidence rectification under complex environmental interference and its benefit for multimodal semantic alignment. The source code is available at <https://github.com/deafenken/UAPA-HCF>.

CCS Concepts

• **Information systems** → **Multimedia and multimodal retrieval**; **Multimedia information systems**; *Video search*; • **Computing methodologies** → *Neural networks*.

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Keywords

Violence Detection, Aleatoric Uncertainty, Prototype Learning, Multimodal Fusion, Weakly Supervised Learning

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1 Introduction

With the widespread deployment of surveillance cameras and the rapid growth of internet video content, developing systems capable of automatically detecting violent incidents holds significant research and practical value [23, 30]. Unlike video classification tasks that typically involve clips containing single events, real-world scenarios often require temporally localizing violent incidents in untrimmed long-form videos [27]. However, obtaining frame-level annotations across massive datasets is prohibitively costly, hindering scalable implementation. Consequently, weakly supervised video violence detection (WS-VAD) has emerged as a more feasible research paradigm: models learn to localize and identify violent segments using only video-level binary labels (normal vs. violent) [8, 30, 32].

Early approaches heavily relied on visual features [12, 21, 30], yet real-world violent behavior often exhibits multimodal characteristics. Critical evidence may be distributed across different modality channels. For instance, key cues for gunshots are primarily conveyed by audio, while physical altercations rely more heavily on motion cues [9, 33, 38]. Recent studies demonstrate that jointly modeling RGB, audio, and optical flow while fully leveraging their complementarity can significantly enhance detection performance [28, 38, 39, 44, 48]. Existing mainstream approaches typically employ attention mechanisms or Transformer-based architectures for cross-modal fusion [16, 34, 36]. However, they often implicitly assume that each modality provides reliable “evidence” in most clips,



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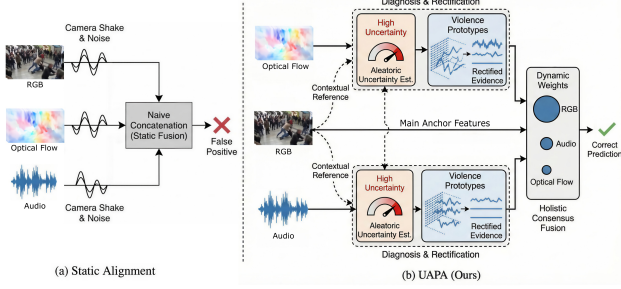


Figure 1: Comparison of fusion paradigms under environmental interference. (a) **Static Alignment** relies on **Naive Concatenation (Static Fusion)**, creating a “hard coupling” where modality-specific noise (e.g., wind in Audio) irreversibly contaminates the joint feature space, leading to **False Positives**. (b) Our **UAPA-HCF** framework employs an **Evidential Rectification** mechanism: RGB serves as a stable **Main Anchor** to estimate **Aleatoric Uncertainty** for auxiliary streams (Audio/Flow); high-uncertainty signals are suppressed and rectified via **Violence Prototypes** and visual context before fusion, ensuring **robust consensus** and **Correct Prediction**.

thereby strengthening multimodal representations via aggregation and weighting.

However, in real-world deployment, a key challenge arises from modality-specific noise. In unconstrained environments, information density varies significantly across modalities and can be easily affected by background interference. For instance, in non-violent scenarios, high-entropy background sounds (such as wind noise) or camera shake may exhibit substantial amplitude fluctuations. Existing methods often misinterpret these as violent cues, leading to false positives [11, 41, 47]. We further observe that traditional fusion pipelines [31, 38] relying on deterministic alignment are particularly sensitive to such modality-level contamination: as shown in Fig. 1(a), when a single modality fails or is dominated by noise in a local segment, simple concatenation or direct aggregation introduces erroneous information into the joint representation, inevitably biasing the overall representation.

To mitigate these issues, we advocate revisiting robust multimodal fusion designs: systems should not directly aggregate multimodal features but instead perform evidence correction prior to fusion. As illustrated in Fig. 1(b), we introduce a decoupling stage before fusion to independently evaluate the reliability of each modality, thereby suppressing unreliable evidence while preserving valid cues. Based on this approach, we propose the UAPA-HCF framework. Unlike previous methods that focus on complex temporal modeling [18, 45], UAPA-HCF redefines fusion as an uncertainty-guided correction process. Specifically, we design the UAPA module: it maintains a learnable “violent prototype” dictionary to capture the normative semantic representation of violent events, and simultaneously uses the RGB stream as a relatively stable anchor to estimate segment-level uncertainty in auxiliary modalities, enabling adaptive suppression of noisy features and alignment to the prototype space. Finally, the HCF module dynamically aggregates

the corrected multimodal streams based on a credibility consensus to output the final violence score.

Our main contributions are summarized as follows:

- We identify *modality heterogeneity*—the imbalance in information density and noise susceptibility across RGB, audio, and optical flow—as the root cause of cross-modal contamination, and formalize a new paradigm: *evidential rectification before fusion*.
- We propose UAPA-HCF with two core innovations: (1) UAPA leverages RGB as a stable anchor to estimate aleatoric uncertainty for auxiliary modalities and performs dynamic interpolation between raw observations and violence prototypes for adaptive rectification; (2) HCF decouples reliability assessment from importance weighting to achieve consensus-based fusion.
- Experiments on XD-Violence show our method achieves state-of-the-art performance among directly comparable multimodal methods with 87.65% AP, while ablation studies and temporal dynamics analysis support the non-redundant contribution of each component and adaptive noise suppression under diverse interference.

2 Related Work

2.1 Multimodal Weakly Supervised Video Violence Detection

Video violence detection has evolved from early benchmark-driven recognition [6] and reconstruction-based anomaly detection [12, 21] to weakly supervised learning with video-level labels. Sultani et al. [30] introduced the MIL paradigm, after which graph modeling [46], discriminative representation learning [35], and temporal feature magnitude learning [32] became common. Recent methods further exploit strong backbones and vision-language or Transformer representations [3, 7, 15, 18, 20, 37, 40].

These studies mainly improve temporal localization by selecting more discriminative snippets, strengthening long-range context, or transferring semantic knowledge from large pre-trained models. They have substantially advanced weakly supervised video anomaly detection, but most of them are still designed around a single visual stream or treat auxiliary cues as additional features after the anomaly representation has already been formed. Consequently, they provide limited guidance on how to handle short-term conflicts among RGB, audio, and motion evidence, which are common in unconstrained violence videos.

Unimodal systems remain fragile in complex scenes, motivating multimodal fusion [5]. XD-Violence [38] highlighted the value of audio, while later methods used optical-flow attention [41], cross-modal distillation [44], multi-scale Transformers [31], hyperbolic embeddings [48], and motion-aware fusion [16]. Yet many pipelines still rely on deterministic aggregation and do not explicitly model snippet-level modality reliability [2, 28, 36]. UAPA-HCF addresses this gap through uncertainty-guided evidence correction before consensus fusion.

More specifically, existing multimodal approaches usually enhance the interaction operator, such as attention, alignment, or score-level weighting, after features from different streams have

been extracted. This design is effective when modalities are mutually supportive, but it may also propagate corrupted auxiliary evidence into the shared representation. Our work is complementary to these fusion architectures: rather than replacing temporal modeling or cross-modal attention, UAPA-HCF inserts a reliability-aware rectification stage that decides whether an auxiliary observation should be trusted, suppressed, or reconstructed before it participates in fusion.

2.2 Uncertainty and Prototype Learning

Uncertainty learning quantifies prediction credibility [1], often through epistemic and aleatoric components [17]. It has improved robustness in evidential learning [26], trusted multi-view learning [11], and video anomaly detection [47]. However, modality-level perceptual noise in multimodal violence detection remains underexplored, especially when one stream is locally corrupted.

Prototype and memory networks store representative event patterns for anomaly discrimination [10, 22, 24]. Prototype learning and margin constraints can further improve feature separability [19, 29, 42]. Unlike memory-reconstruction methods, UAPA learns updatable violence prototypes as semantic anchors and combines them with credibility estimation so that unreliable evidence is rectified before fusion [14].

3 Methodology

To address cross-modal interference under weak supervision, we propose the UAPA-HCF framework. As shown in Fig. 2, our architecture consists of three core components: (1) modality-specific feature extraction with explicit projection formulas, (2) the Uncertainty-Aware Prototype Anchoring (UAPA) module for evidential rectification, and (3) the Holistic Consensus Fusion (HCF) module for robust aggregation. Below we detail each component’s design and mathematical formulation.

3.1 Feature Extraction and Unified Projection

Given an untrimmed video V , we divide it into a bag of T non-overlapping snippets $\{v_t\}_{t=1}^T$. For each snippet, we extract features from three distinct channels using frozen pre-trained backbones:

RGB vision: Extracted via I3D [4] pre-trained on Kinetics-400:

$$Z_v^{(t)} = \mathcal{F}_{\text{I3D}}(v_t) \in \mathbb{R}^{D_v}, \quad D_v = 1024 \quad (1)$$

Audio: Spectrogram features extracted via VGGish [13] pre-trained on AudioSet:

$$Z_a^{(t)} = \mathcal{F}_{\text{VGGish}}(\text{Spectrogram}(v_t)) \in \mathbb{R}^{D_a}, \quad D_a = 128 \quad (2)$$

Optical flow: Motion features extracted via I3D-Flow:

$$Z_f^{(t)} = \mathcal{F}_{\text{I3D-Flow}}(\text{Flow}(v_t)) \in \mathbb{R}^{D_f}, \quad D_f = 1024 \quad (3)$$

Since raw features originate from heterogeneous manifolds with varying dimensions, we project them into a unified semantic hyperspace $\mathbb{R}^D$ ($D = 128$) via modality-specific fully connected layers with ReLU activation and dropout ($p = 0.1$):

$$\tilde{Z}_m^{(t)} = \text{Dropout}\left(\delta\left(\mathbf{W}_m Z_m^{(t)} + \mathbf{b}_m\right)\right), \quad m \in \{v, a, f\} \quad (4)$$

where $\delta(\cdot)$ denotes the ReLU activation function. We designate the RGB stream as the Main Anchor ($Z_m = \tilde{Z}_v$) due to its superior semantic stability and lower susceptibility to transient environmental noise. Audio and optical flow are treated as Auxiliary Modalities with distinct variables $Z_{\text{aux}}^a = \tilde{Z}_a$ and $Z_{\text{aux}}^f = \tilde{Z}_f$, whose reliability must be dynamically assessed before integration.

3.2 Preliminaries: Heterogeneity of Multimodal Evidence

Weakly supervised violence detection assumes that violent moments generate discriminative signatures across modalities. However, RGB, audio, and optical flow exhibit profound heterogeneity in information density and noise susceptibility.

RGB features provide stable semantic grounding with moderate entropy but suffer from visual ambiguity under occlusions or poor lighting. Audio streams deliver bursty, high-magnitude responses for violent events (e.g., gunshots, screams), yet are frequently corrupted by environmental noise (wind, traffic) with similar spectral characteristics—yielding the strongest discriminative power alongside the highest false-alarm risk. Optical flow captures motion dynamics essential for physical interactions but is highly vulnerable to camera shake; unlike audio noise, flow artifacts are structurally indistinguishable from genuine violence signatures, making them particularly insidious.

This creates a critical dilemma: the most informative modalities (audio for explosions, flow for fights) are also the most noise-prone. Traditional fusion methods implicitly assume uniform reliability across modalities and time. When high-entropy auxiliary signals coincide with benign visual content, naive concatenation forces contradictory evidence into joint representations, inevitably elevating anomaly scores for non-violent segments.

To quantify static reference statistics that prove inadequate for dynamic reliability assessment, we define the global mean feature representation for any modality m as:

$$\mu_m = \frac{1}{T} \sum_{t=1}^T Z_m^{(t)} \quad (5)$$

However, as violent events generate highly scenario-dependent auxiliary signatures—a gunshot yields markedly different audio patterns than a physical altercation—global statistics such as μ_m cannot serve as reliable references for snippet-level reliability estimation.

3.3 Uncertainty-Aware Prototype Anchoring

Cross-modal interference arises from a fundamental mismatch: fusion mechanisms assume uniformly reliable evidence, yet auxiliary modalities frequently violate this assumption in unconstrained environments. Instead of post-hoc denoising, we propose evidential rectification that explicitly assesses snippet-level reliability before fusion. UAPA implements this through three principles: (1) reliability must be estimated relative to a stable reference; (2) unreliable evidence should be suppressed rather than forced into alignment; (3) semantic consistency should be enforced through canonical violence patterns.

Anchor-guided contextual interaction Assessing auxiliary modality reliability demands a stable reference that reflects the

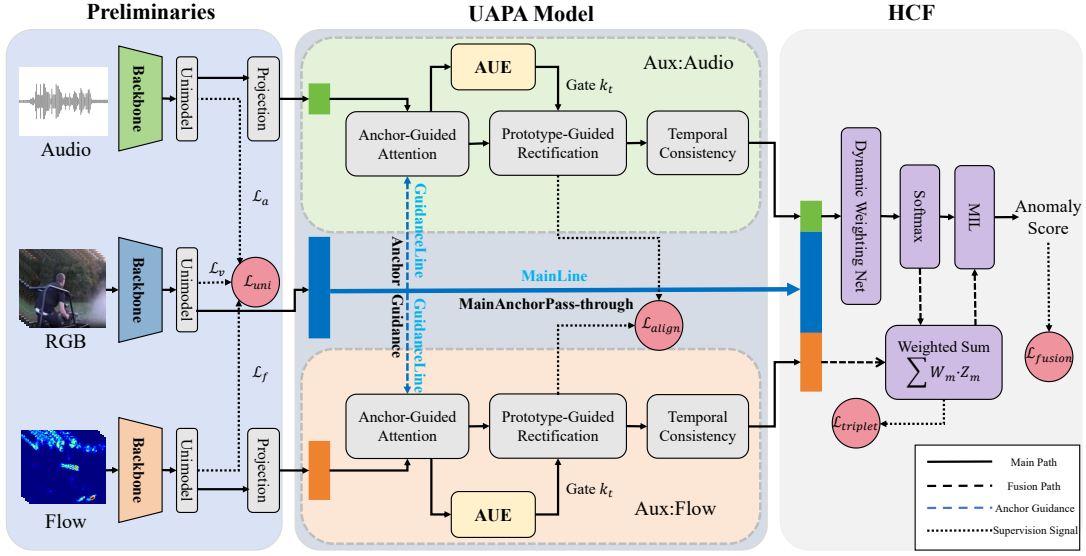


Figure 2: Overview of the proposed UAPA-HCF framework. It consists of three main stages: (1) Feature Extraction for RGB, Audio, and Flow; (2) The UAPA Module (center dashed box), which estimates instance-level uncertainty κ_t to dynamically weigh between prototype anchoring and context compensation; and (3) Holistic Consensus Fusion (HCF) to generate the final anomaly score.

expected auxiliary representation conditioned on the current visual context. We leverage the semantic stability of RGB streams to generate a context-aware reference through cross-modal attention:

$$Q_m = Z_{\text{aux}}^m \mathbf{W}_Q^m, \quad K = Z_m \mathbf{W}_K, \quad V = Z_m \mathbf{W}_V, \quad m \in \{a, f\} \quad (6)$$

$$H_{\text{inter}}^m = \text{Attention}(Q_m, K, V) = \text{softmax} \left(\frac{Q_m(K)^T}{\sqrt{d_k}} \right) V \quad (7)$$

The resulting H_{inter}^m represents the auxiliary feature inferred purely from visual context, serving as a dynamic reference standard. Alignment between the observed Z_{aux}^m and H_{inter}^m indicates coherence with visual semantics—screams naturally accompanying physical struggle, for instance—while significant divergence signals potential environmental corruption such as wind noise intruding upon an otherwise calm visual scene.

Aleatoric uncertainty estimation The semantic discrepancy between Z_{aux}^m and H_{inter}^m provides a natural signal for modeling aleatoric uncertainty—noise inherent to the data acquisition process rather than model limitations. Because this discrepancy arises from environmental conditions during data collection (e.g., wind affecting microphones, camera shake affecting motion sensors), it must be explicitly quantified to prevent noise propagation. We map the joint representation $[Z_{\text{aux}}^m; H_{\text{inter}}^m]$ through a non-linear perception network to produce a continuous confidence gate $\kappa_t^{(m)} \in (0, 1)$:

$$\kappa_t^{(m)} = \sigma \left(\mathbf{w}_2^{(m)T} \delta \left(\mathbf{W}_1^{(m)} [Z_{\text{aux}}^m; H_{\text{inter}}^m] + \mathbf{b}_1^{(m)} \right) \right) \quad (8)$$

where $\sigma(\cdot)$ denotes the sigmoid function, $\delta(\cdot)$ denotes ReLU activation, and $[\cdot; \cdot]$ indicates concatenation. The sigmoid output enables soft gating: values approaching 1 reflect high confidence where auxiliary signals provide reliable complementary evidence, while values near 0 indicate dominant environmental interference. This continuous formulation delivers three advantages over binary gating strategies: uninterrupted gradient flow during training, retention of partially reliable cues from moderately degraded signals, and interpretability of the gate itself as a diagnostic indicator of modality trustworthiness.

Prototype-guided rectification Confidence gating alone remains insufficient; outright suppression of uncertain features discards potentially informative content. We therefore introduce semantic rectification, which reconstructs unreliable auxiliary features using context from the stable anchor modality, regularized by canonical violence patterns. A learnable dictionary of Canonical Violence Prototypes $\mathbf{P} \in \mathbb{R}^{K \times D}$ ($K = 64$) defines semantic attractors that shape the feature subspace toward violence-relevant manifolds.

We define the projection operator $\phi(\cdot)$ that anchors features to the nearest prototype in Euclidean space:

$$\phi(z) = z - (z - \mathbf{p}^*), \quad \text{where } \mathbf{p}^* = \arg \min_{\mathbf{p} \in \mathbf{P}} \|z - \mathbf{p}\|_2 \quad (9)$$

The rectification process performs dynamic interpolation between raw evidence and context-compensated representation:

$$Z_{\text{sub}}^m = \kappa_t^{(m)} \cdot \phi(Z_{\text{aux}}^m) + (1 - \kappa_t^{(m)}) \cdot \phi(H_{\text{inter}}^m), \quad m \in \{a, f\} \quad (10)$$

When confidence is high ($\kappa_t^{(m)} \rightarrow 1$, corresponding to low aleatoric uncertainty), the model anchors the observed auxiliary feature to the nearest violence prototype via $\phi(\cdot)$, enhancing discriminability and semantic alignment. Under low confidence (high aleatoric uncertainty, $\kappa_t^{(m)} \rightarrow 0$), the corrupted observation is suppressed and replaced by H_{inter}^m —the visually inferred auxiliary representation that remains stable despite signal degradation. The rectified feature Z_{sub}^m is subsequently projected back to dimension D to yield $Z_{\text{aux}}^{m'}$.

Temporal consistency and normalization Environmental noise often manifests as transient bursts such as a single wind gust, while genuine violence typically spans multiple consecutive snippets. To enforce temporal smoothness on the rectified representation, we apply a depth-wise temporal convolution that aggregates local context:

$$\hat{Z}_{\text{aux}}^{m'} = Z_{\text{aux}}^{m'} + \text{DWConv1d}(Z_{\text{aux}}^{m'}) \quad (11)$$

After temporal smoothing, we apply layer normalization to balance feature magnitudes across modalities and stabilize training dynamics:

$$\bar{Z}_{\text{aux}}^{m'} = \text{LayerNorm}(\hat{Z}_{\text{aux}}^{m'}) \quad (12)$$

where $\text{LayerNorm}(x) = \gamma \frac{x - \mu}{\sigma} + \beta$ with μ and σ computed across the feature dimension.

Multimodal feature concatenation After denoising and normalization, the three modalities are concatenated into a unified representation for downstream fusion:

$$Z_{\text{cat}}^{(t)} = [Z_m^{(t)}; \bar{Z}_{\text{aux}}^{a'}(t); \bar{Z}_{\text{aux}}^{f'}(t)] \in \mathbb{R}^{3D} \quad (13)$$

This concatenation preserves the distinct semantic spaces of each modality while enabling holistic interaction in the subsequent fusion stage.

3.4 Holistic Consensus Fusion

Even after rectification, modalities retain complementary strengths: audio excels at detecting acoustic violence including explosions and gunshots, while flow better captures motion-intensive events such as fights and riots. Rather than assigning fixed importance weights, HCF implements a dynamic consensus mechanism that determines modality contribution based on snippet-specific credibility.

Dynamic consensus weighting Modality importance is event-dependent and time-varying. A silent physical altercation demands heavy reliance on flow features, while a distant explosion may be primarily audible. To model this adaptivity, we employ modality-specific scoring networks $\mathcal{G}_m(\cdot)$ that predict a scalar energy score $E_t^{(m)}$ reflecting each stream’s discriminative power at timestamp t .

Each scoring network $\mathcal{G}_m$ is implemented as a two-layer MLP with ReLU activation. Given the rectified feature $Z_m^{(t)} \in \mathbb{R}^D$ of modality m at time t , we first project it to a hidden representation:

$$h_t^{(m)} = \delta(\mathbf{W}_m^{(1)} Z_m^{(t)} + \mathbf{b}_m^{(1)}), \quad \mathbf{W}_m^{(1)} \in \mathbb{R}^{D_h \times D} \quad (14)$$

where $\delta(\cdot)$ denotes ReLU activation and $D_h = D/4 = 32$ is the hidden dimension (reduction ratio $r = 4$). The energy score is then computed as a linear projection to a scalar:

$$E_t^{(m)} = \mathbf{w}_m^{(2)\top} h_t^{(m)} + b_m^{(2)}, \quad \mathbf{w}_m^{(2)} \in \mathbb{R}^{D_h} \quad (15)$$

Crucially, each modality $m \in \{v, a, f\}$ uses independent parameters $\{\mathbf{W}_m^{(1)}, \mathbf{b}_m^{(1)}, \mathbf{w}_m^{(2)}, b_m^{(2)}\}$ to capture their distinct statistical properties—audio energy distributions differ fundamentally from visual feature magnitudes, and a shared scorer would conflate these modalities’ intrinsic scales.

These energy scores are normalized via softmax to produce dynamic attention weights that sum to one:

$$\alpha_t^{(m)} = \frac{\exp(E_t^{(m)})}{\sum_{k \in \{v, a, f\}} \exp(E_t^{(k)})}, \quad \sum_m \alpha_t^{(m)} = 1 \quad (16)$$

The final consensus representation aggregates the rectified features according to their instantaneous credibility:

$$Z_{\text{HCF}}^{(t)} = \sum_{m \in \{v, a, f\}} \alpha_t^{(m)} \cdot Z_m^{(t)} \quad (17)$$

This formulation ensures that at each timestamp, the fused representation is dominated by the most reliable evidence—whether visual, acoustic, or motion-based—while still incorporating supporting signals from other modalities with reduced weights. Unlike static fusion or vanilla attention, HCF explicitly decouples reliability assessment (handled by UAPA) from importance weighting (handled by HCF), creating a two-stage robustness pipeline.

3.5 Loss Functions

The training objective comprises four complementary loss terms to ensure both modality alignment and accurate violence detection. The total loss $\mathcal{L}_{\text{total}}$ is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{uni}} + \lambda_1 \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{fusion}} + \lambda_3 \mathcal{L}_{\text{triplet}} \quad (18)$$

where $\lambda_1 = 10$, $\lambda_2 = 10$, and $\lambda_3 = 0.001$ are hyperparameters balancing the contributions of each term. Each component is defined as follows:

- (1) **Unimodal MIL Loss ($\mathcal{L}_{\text{uni}}$):** Supervises individual modality branches during initial training stages. For each modality m , we compute the video-level anomaly score as the maximum snippet score:

$$S_m = \max_{t \in \{1, \dots, T\}} \mathbf{w}_m^\top Z_m^{(t)} \quad (19)$$

The loss is binary cross-entropy between S_m and the video-level label $y \in \{0, 1\}$:

$$\mathcal{L}_{\text{uni}} = \sum_{m \in \{v, a, f\}} [-y \log(S_m) - (1 - y) \log(1 - S_m)] \quad (20)$$

- (2) **Alignment Loss ($\mathcal{L}_{\text{align}}$):** Enforces semantic consistency between the main (RGB) and auxiliary modalities. It comprises three sub-components:

- **Feature distance:** Cosine similarity loss between RGB features and context-inferred auxiliary representations:

$$\mathcal{L}_{\text{dist}} = \sum_{m \in \{a, f\}} \left(1 - \frac{Z_m^\top H_{\text{inter}}^m}{\|Z_m\|_2 \|H_{\text{inter}}^m\|_2} \right) \quad (21)$$

- Temporal consistency: Constraint ensuring aligned features maintain temporal smoothness:

$$\mathcal{L}_{\text{temp}} = \sum_{m \in \{a, f\}} \sum_{t=2}^T \|Z_{\text{aux}}^{m(t)} - Z_{\text{aux}}^{m(t-1)}\|_2^2 \quad (22)$$

- Auxiliary classification: Cross-entropy losses on auxiliary branches:

$$\mathcal{L}_{\text{cls}} = \sum_{m \in \{a, f\}} [-y \log(S_m) - (1 - y) \log(1 - S_m)] \quad (23)$$

The complete alignment loss is:

$$\mathcal{L}_{\text{align}} = \mathcal{L}_{\text{dist}} + \mathcal{L}_{\text{temp}} + \mathcal{L}_{\text{cls}} \quad (24)$$

- (3) Fusion MIL Loss ($\mathcal{L}_{\text{fusion}}$): Standard Multiple Instance Learning loss applied to the final fused representation:

$$S_{\text{fusion}} = \max_{t \in \{1, \dots, T\}} \mathbf{w}_{\text{fusion}}^\top Z_{\text{HCF}}^{(t)} \quad (25)$$

$$\mathcal{L}_{\text{fusion}} = -y \log(S_{\text{fusion}}) - (1 - y) \log(1 - S_{\text{fusion}}) \quad (26)$$

- (4) Triplet Loss ($\mathcal{L}_{\text{triplet}}$): Margin-based triplet loss optimizing the embedding space geometry. Let $\mathbf{c}_n$ denote the centroid of normal snippets, $\mathbf{z}_p^{(t)}$ a violent (positive) snippet feature, and $\mathbf{z}_n^{(t)}$ a normal (negative) snippet feature. The loss pulls violent snippets away from normal centroids while pushing normal snippets toward their centroid:

$$\mathcal{L}_{\text{triplet}} = \frac{1}{|\mathcal{P}|} \sum_{t \in \mathcal{P}} \max \left(\|\mathbf{z}_n^{(t)} - \mathbf{c}_n\|_2^2 - \|\mathbf{z}_p^{(t)} - \mathbf{c}_n\|_2^2 + \alpha, 0 \right) \quad (27)$$

where $\mathcal{P}$ is the set of violent snippets in the video and $\alpha = 0.2$ is the margin hyperparameter.

4 Experiments

4.1 Datasets and Evaluation Metrics

We evaluate the proposed framework on XD-Violence [38], a large-scale benchmark for multimodal violence detection. It contains 4,754 untrimmed videos (217 hours) collected from movies and in-the-wild web videos, covering six categories: Abuse, Car Accident, Explosion, Fighting, Riot, and Shooting. The dataset includes 2,405 violent and 2,349 non-violent videos, and follows the standard split [38] with 3,954 training videos and 800 testing videos. The benchmark is challenging due to diverse audiovisual correlations and frequent environmental noise.

Following prior work [30, 32, 38], we use frame-level Average Precision (AP) as the primary metric. Because violent events are sparse in untrimmed videos, AP better reflects ranking quality over time compared to accuracy or AUC.

4.2 Implementation Details

For feature extraction, we utilize off-the-shelf pre-trained models without further fine-tuning to extract unimodal representations. For vision, we adopt I3D [4] pre-trained on Kinetics-400 and extract RGB and Flow features from 16-frame non-overlapping snippets. For audio, we use VGGish [13] pre-trained on AudioSet. All modalities are projected to $D=128$ via modality-specific FC layers with ReLU and dropout ($p=0.1$). In UAPA, the prototype dictionary size

is $K=64$, and the anchor-guided attention uses 4 heads with head dimension 32. In HCF, we set the reduction ratio to $r=4$. The classifier is a three-layer MLP with sigmoid output.

We implement the model in PyTorch and train on a single NVIDIA RTX 3090 GPU. We use Adam with learning rate $1e^{-4}$, batch size 2, weight decay $5e^{-4}$, and train for 1000 epochs. Hyperparameters are set to $\lambda_1=10$, $\lambda_2=10$, and $\lambda_3=0.001$ based on grid search, and we fix the random seed to 2026 for reproducibility.

4.3 Computational Complexity

UAPA-HCF operates on pre-extracted snippet-level features rather than raw frames, so the dominant cost in deployment remains the backbone feature extraction stage shared by most multimodal baselines. The extra modules are anchor-guided attention, uncertainty gates, prototype lookup, and small modality scoring MLPs. For a video with T snippets, feature dimension D , and K prototypes, the attention step costs $O(T^2D)$ per auxiliary modality, while prototype assignment costs $O(TKD)$. Since we use $D=128$, $K=64$, and a reduced hidden dimension $D/4$ in HCF, the fusion modules add modest feature-level computation compared with dense video backbone inference. This design is therefore intended for scenarios where RGB/audio/flow features can be extracted offline or streamed from existing surveillance pipelines, while more detailed real-time deployment optimization remains future work.

4.4 Comparison with State-of-the-Art Methods

We compare UAPA-HCF with representative state-of-the-art methods on XD-Violence in Table 1. Following common practice, we report frame-level AP and group methods by their modality usage. To keep the comparison fair, the table focuses on methods with comparable XD-Violence evaluation protocols and modality settings. We do not directly merge closed-source methods or methods relying on substantially different auxiliary supervision/protocols unless their reported setting is sufficiently aligned for direct comparison.

As shown in Table 1, UAPA-HCF achieves state-of-the-art performance among directly comparable methods, reaching 87.65% AP in the tri-modal setting. Compared with the strongest aligned tri-modal baseline, Jin et al. [16], our method improves AP from 86.07% to 87.65% (+1.58). The gain supports our main claim: explicitly estimating aleatoric uncertainty and rectifying evidence before fusion prevents noisy auxiliary observations from being forced into the shared representation.

The modality variants further show that the improvement is not due to simple capacity growth. RGB-only UAPA-HCF reaches 79.12% AP, audio raises performance to 85.47%, flow raises it to 83.76%, and combining audio and flow reaches 87.65%. Thus acoustic and motion cues are complementary, but their benefit is strongest when the model can suppress unreliable segments and let the fused decision be dominated by credible evidence.

4.5 Ablation Study

Table 2 isolates the architectural contribution of each component. The plain concatenation baseline reaches 79.62% AP, indicating that direct aggregation of heterogeneous streams remains vulnerable to noisy auxiliary evidence. Adding prototype anchoring increases

Table 1: Comparison on XD-Violence. Methods are grouped by modality; the best and second-best directly comparable results are bolded and underlined. “V” denotes RGB without optical flow.

Method	Modality	AP (%)
<i>Semi-supervised Anomaly Detection Methods</i>		
SVM baseline	V	50.78
OCSVM [25]	V	27.25
Hasan et al. [12]	V	30.77
<i>Weakly Supervised Video Anomaly Detection Methods</i>		
Sultani et al. [30]	V	73.20
Wu et al. (RGB only) [38]	V	75.41
RTFM [32]	V	77.81
MSL [18]	V	<u>78.28</u>
Wu et al. (RGB + Audio) [38]	V+A	78.64
AVVD [39]	V+A	78.64
UR-DMU [47]	V+A	81.77
MACIL-SD [44]	V+A	83.40
MCL [43]	V+A	84.03
HyperVD [48]	V+A	85.67
Wu et al. (RGB + Audio + Flow) [38]	V+A+F	79.53
OAMFN [41]	V+A+F	83.09
MSBT [31]	V+A+F	84.32
Jin et al. [16]	V+A+F	<u>86.07</u>
UAPA-HCF (Ours)	V	79.12
UAPA-HCF (Ours)	<u>V+A</u>	<u>85.47</u>
UAPA-HCF (Ours)	V+F	83.76
UAPA-HCF (Ours)	V+A+F	87.65

Table 2: Ablation study of components in UAPA-HCF.

Prototypes	Uncertainty (AUE)	Consensus Fusion	AP (%)
-	-	Concat	79.62
✓	-	Concat	84.80
✓	✓	Concat	86.85
✓	✓	HCF (Ours)	87.65

AP to 84.80%, suggesting that canonical violence prototypes provide a useful semantic reference for rectifying ambiguous features. Introducing uncertainty estimation further improves AP to 86.85%, confirming that reliability-aware gating is important for deciding when auxiliary evidence should be trusted or compensated. Finally, replacing concatenation with HCF yields 87.65% AP, showing that consensus-based weighting complements UAPA by adapting each modality’s contribution after feature rectification.

Table 3 reports a leave-one-out ablation of the training objective.

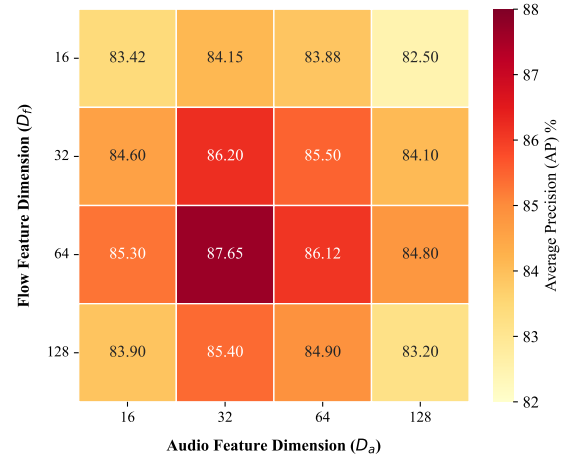
The full objective (Row 5) achieves the best AP of 87.65%, and removing any term consistently hurts performance. $\mathcal{L}_{uni}$ provides branch-wise supervision, $\mathcal{L}_{align}$ keeps auxiliary streams coherent with the RGB anchor, $\mathcal{L}_{fusion}$ directly optimizes the HCF output,

Table 3: Leave-one-out ablation of training losses.

Index	$\mathcal{L}_{uni}$	$\mathcal{L}_{align}$	$\mathcal{L}_{fusion}$	$\mathcal{L}_{triplet}$	AP (%)
1	✓	✓	✓	-	83.21
2	✓	✓	-	✓	84.87
3	✓	-	✓	✓	84.23
4	-	✓	✓	✓	86.78
5	✓	✓	✓	✓	87.65

Table 4: Sensitivity to the number of violence prototypes K .

K	16	32	64	128
AP (%)	83.27	86.94	87.65	87.42

**Figure 3: Sensitivity to audio and optical-flow projection dimensions. The best AP of 87.65% is obtained at $D_a = 32$, $D_f = 64$.**

and $\mathcal{L}_{triplet}$ shapes a more separable embedding space. The large drops after removing $\mathcal{L}_{align}$ or $\mathcal{L}_{triplet}$ indicate that semantic consistency and metric regularization are especially important for the “rectify-before-fuse” design.

We further analyze the prototype dictionary size in Table 4. AP increases from 83.27% to 87.65% as K grows from 16 to 64, showing that too few prototypes under-cover the diverse semantics of violent events. Increasing K to 128 slightly reduces AP to 87.42%, suggesting that redundant prototypes bring limited additional capacity and may introduce noisier matches under weak supervision. We therefore set $K=64$ as a compact default that balances semantic coverage and robustness.

Furthermore, we study how the projection dimensions of audio (D_a) and flow (D_f) affect performance. As illustrated in Fig. 3, AP follows an inverted U-shape trend and reaches the best value (87.65%) at $D_a=32$ and $D_f=64$. Very small dimensions limit semantic capacity, while overly large dimensions introduce redundancy and can amplify noise, which is particularly harmful under weak

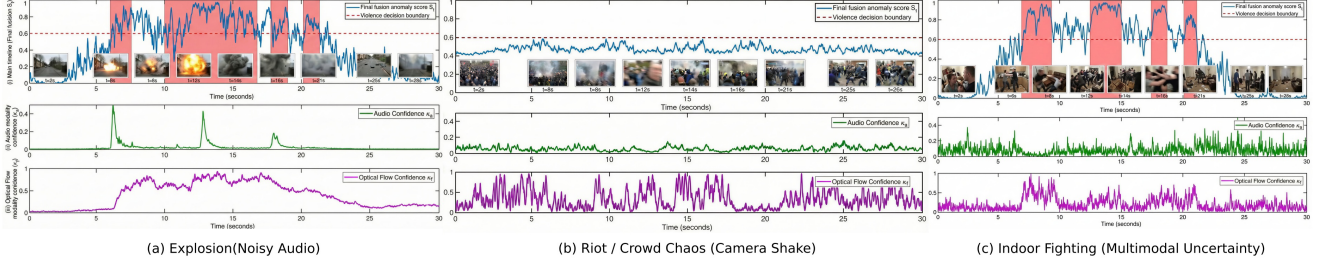


Figure 5: Temporal dynamics under environmental interference. The plots show anomaly score S_t , audio confidence $\kappa_t^{(a)}$, and optical-flow confidence $\kappa_t^{(f)}$ for benign audio burst, camera shake, and indoor occlusion cases.

supervision. Fig. 3 also suggests that flow benefits from a larger dimension than audio (64 vs. 32), consistent with the intuition that motion patterns are often more information-dense.

4.6 Qualitative Analysis

We visualize t-SNE embeddings of test samples in Fig. 4. Raw features show substantial overlap between normal and violent samples, whereas features processed by UAPA-HCF form more compact clusters with clearer margins. This suggests that prototype anchoring and uncertainty-based rectification improve both intra-class compactness and inter-class separability.

Beyond static distributions, Fig. 5 examines whether UAPA adapts over time under typical environmental interference. The left column contains a benign audio burst: the raw acoustic response is high, but the audio confidence decreases and the fused score stays below the decision threshold. This indicates that UAPA does not simply treat high-energy audio as violence; instead, the gate compares the auxiliary signal with RGB-guided context and learned violence prototypes. The resulting score remains temporally consistent with the ground-truth interval while gate signals adapt to local disturbances and modality-specific volatility.

In the middle column, riot footage with strong camera shake creates chaotic motion between 15–22 seconds, a pattern that can easily mislead flow-dependent baselines. UAPA suppresses optical-flow confidence ($\kappa_t^{(f)} \rightarrow 0$) while maintaining high audio confidence ($\kappa_t^{(a)} \approx 0.9$) for crowd screams. This contrast shows that uncertainty estimation relies on contextual coherence rather than absolute feature magnitude. The fused score consequently tracks

true violence while ignoring shake-induced flow artifacts, which is crucial for surveillance scenarios where abrupt ego-motion can resemble physical confrontation.

The right column further illustrates dynamic modality handover. During indoor fighting, no single modality remains consistently reliable. Under visual occlusion around 10–15 seconds, flow confidence rises ($\kappa_t^{(f)} \approx 0.85$) to compensate for ambiguous RGB; during vocal struggles around 20–25 seconds, audio becomes dominant while flow recedes. Despite these confidence fluctuations, S_t stays above the threshold τ throughout the violent segment, showing that the final prediction does not collapse when one modality becomes uncertain. These dynamics provide direct evidence that UAPA-HCF rectifies environmental noise at the gate level and then weights modalities according to contextual reliability, turning multimodal fusion from a potential source of contamination into a robustness amplifier under real-world interference.

5 Conclusion

We presented UAPA-HCF, which rectifies multimodal evidence through uncertainty-aware prototype anchoring and synthesizes it via holistic consensus fusion. By preventing noisy modalities from contaminating fused representations, UAPA-HCF achieves state-of-the-art performance among directly comparable multimodal methods on XD-Violence and improves resistance to modality-specific noise. Future work will extend this rectification-before-fusion principle to broader multimodal retrieval tasks where modality reliability varies over time.

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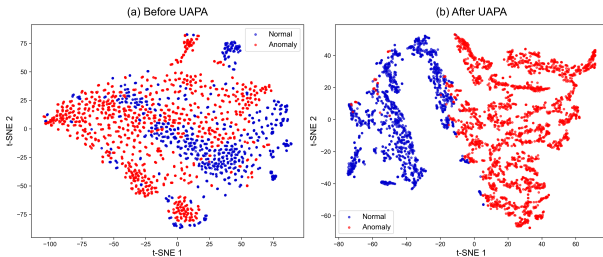


Figure 4: t-SNE visualization. (a) Raw features. (b) Features processed by UAPA-HCF, showing clearer separation.

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